

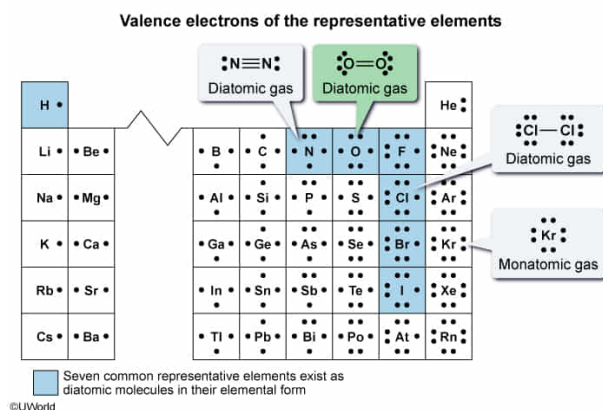
Based on the Lewis model of bonding, an element that exists in its standard state as a diatomic gas and contains a double covalent bond is:

- ☐ A. nitrogen. [7%]
✓ ☒ B. oxygen. [87%]
☐ C. chlorine. [2%]
☐ D. krypton. [2%]

Correct

86% answered correctly

Explanation:



Of the common **representative elements**, seven exist as **diatomic molecules** in their standard **elemental state**. These seven elements are hydrogen, nitrogen, oxygen, fluorine, chlorine, bromine, and iodine. To achieve a more stable electron configuration with a full **valence shell**, atoms of these elements form **covalently bonded pairs** by **sharing** valence electrons and occur naturally as H_2 , N_2 , O_2 , F_2 , Cl_2 , Br_2 , and I_2 , respectively. These elements exist as gases at room temperature with the exception of bromine (a liquid) and iodine (a solid).

Due to differences in orbitals and the number of valence electrons, H_2 , F_2 , Cl_2 , Br_2 , and I_2 can achieve a full valence shell by forming a pair with only a single covalent bond, but O_2 must form a double bond and N_2 must form a triple bond. In the **second period** (row) on the periodic table, nitrogen, oxygen, fluorine, and neon are all elemental gases, but only oxygen exists as a diatomic molecule with a double covalent bond in its elemental state.

(Choice A) Nitrogen is an elemental gas in the second period, but it forms a diatomic molecule with a *triple* covalent bond in its elemental state.

(Choice C) Chlorine is an elemental gas in the *third* period. Also, chlorine forms a diatomic molecule with only a *single* covalent bond in its elemental state.

(Choice D) Krypton is an elemental gas in the *fourth* period. As a noble gas, it does not form a diatomic molecule in its elemental state.

Educational objective:

Of the common representative elements, hydrogen, nitrogen, oxygen, fluorine, chlorine, bromine, and iodine occur naturally as covalently bonded diatomic molecules in an elemental state to achieve a full valence shell. Each of these elements exists as a gas at room temperature except for bromine (a liquid) and iodine (a solid).

Time spent:0 sec

Subject:

General Chemistry

QID:401730

System:

Atomic Theory and Chemical Composition